

Uttkarsh Joshi

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Hiring Team

Amazon Web Services

Applied Scientist I – Speech and Language Technology

Job ID: 10421937 | ADCI – Karnataka

Dear Hiring Team,

I am writing to apply for the Applied Scientist I position within Amazon's Speech and Language Technology team. Machine learning has been the thread running through most of my engineering work over the past two years, from building semantic similarity systems to integrating AI-powered features in production, and I am eager to bring that hands-on experience to a team working at the scale and depth that Amazon operates at.

During my internship at Rensei Inc., I worked on AI-powered backend platform features, gaining direct experience with how machine learning models are integrated into production systems and how their outputs need to be reliable, efficient, and explainable to the people building around them. That experience sharpened my understanding of the full lifecycle of an ML-powered system, from data handling and model integration through to latency constraints and real-world performance.

My most relevant project is a semantic caching and query optimization engine I built in Python, which applies embedding-based vector similarity to match incoming queries against a cached index. This is the same class of technique that underpins modern NLU and retrieval-augmented generation systems. I used cosine similarity matching, NumPy and Pandas for vector operations, and PostgreSQL for analytics, ultimately reducing redundant compute by close to 35% and cutting average response latency from 2.1 seconds to 1.2 seconds. The project gave me practical experience with the tradeoffs involved in deploying similarity-based ML inference at scale, including memory management, eviction strategy, and evaluation methodology.

I am also actively pursuing a professional certification in Artificial Intelligence and Machine Learning and have studied the core architecture of large language models including transformer-based attention mechanisms, training and inference lifecycles, and the kinds of optimizations that matter when serving these models at scale. Natural language understanding and spoken language technology are areas I find genuinely compelling, and I would welcome the opportunity to contribute to systems that reach millions of users.

I am based in India and would be excited to join the Bangalore team. Thank you for considering my application.

Sincerely,
Uttkarsh Joshi